1A6: Simulation of a rocket engine

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Abstract

In this report, we are trying to design a simplified version of a rocket engine. The fuel is Hydrogen gas $H2$, heated up to $T =$ 10000kelvin. Our goal is to make a satellite with 1000 kg to achieve the escape velocity to the outer space.

1 Introduction

Venturing into space is the most important achievements of human being so far. Driven by our curiosity about our position in this vast universe,we have harnessed every possible resource to extend our understanding of the cosmos. the main variable in this understanding is deploying devices to help us collecting data. In this short study, our aim is to find a method to launch a satellite to the outer space and simulate it numerically .

We will generate a random planet in random solar system with information we need like: planet radius,mass are imported from the "ast2000tools" class. We have selected (H2) as a fuel source. By heating it up to $T = 10000$ kelvin and achieve optimal kinetic energy , without causing it to ignite We will make a hole in the fuel tank allowing the heated particles to escape and generating the thrust needed to launch our rocket. for this simulation ,We will use small fuel tanks (as boxes) ,each with side length of 10^{-06}m containing 10^{05}particles. Each of these boxes will contribute to total thrust needed.

Our aim is to achieve the necessary acceleration within a time of 20 minutes which enables the satellite to escape the planet's gravitational pull enter the outer space. To accomplish this , we need to calculate the amount of fuel required for this mission . We will use a simplified version of a space rocket consisting of three main parts :

- The upper part , where we have our satellite.
- The middle part(the combustion chamber) where all the fuel that will be converted into hot (H2) will be stored and heated.
- The lower section ,consists of many small boxes .Each box has a hole that allows the heated (H2) to escape and generate the thrust needed for the launch.

As shown in Figure 1

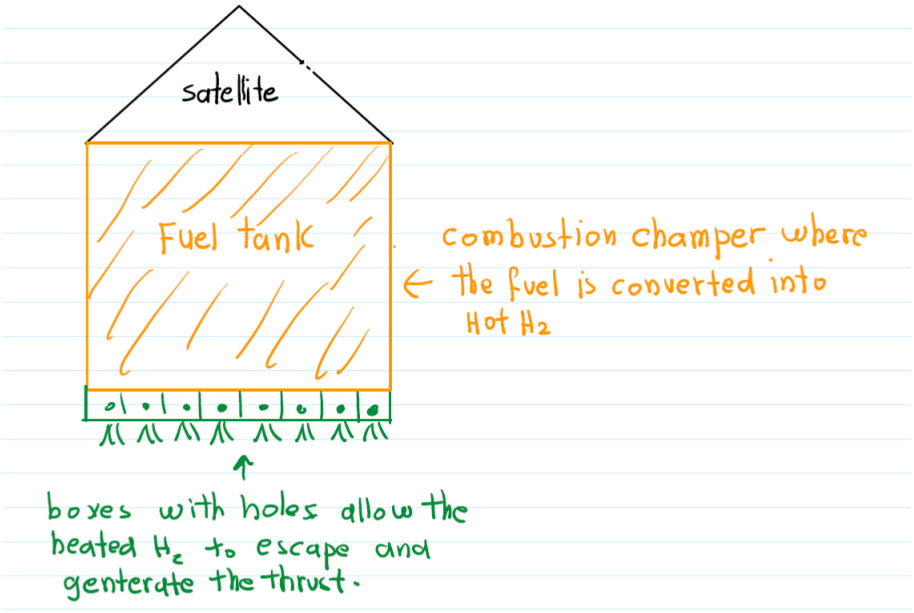


Figure 1: A plot shows a simplified version of space rocket with three major parts

In this simulation some Simplifications were made :

- We assumed that the gas is ideal with no interactions between particles.

- Collisions with the walls are elastic, conserving both kinetic energy and momentum.
- Particle escape is modeled virtually, without physically removing particles from the simulation.

2 Method

Firstly, We begin with the calculation of the escape velocity of our planet, which is given by the formula derived from the conservation of energy:

$$v_{esc} = \sqrt{\frac{2GM}{R}}$$

Where:

- G is the gravitational constant.
- M represents the mass of our planet.
- R is the radius of our planet.

For our planet which has 3.185 Earth mass,the escape velocity was $v_{esc} =$ 16.447Km/s.
Next step is to start designing the engine. we start with a square small box with side length $L = 10^{-}6$m containing 10^5 particles. Their positions were selected randomly with a uniform distribution :$\bar{r}_i = (x_i, y_i, z_i), x_i, y_i, z_i \in [-\frac{L}{2}, \frac{L}{2}]$.
Then we generated random velocities follow the normal distribution with a standard deviation given by:

$$\sigma = \sqrt{\frac{kT}{m}}$$

.

- k is the Boltzmann constant.
- T is the temperature of the gas.
- m is the mass of the hydrogen molecule ($H2$).

. This standard deviation will be used to generate the velocities in the range of σ .
After that, We found the mean values of both positions and velocities. The theoretical formula for the mean velocity as deduced in the exercises of the lecture notes "1B" , is :

$$\bar{v} = \sqrt{\frac{8kT}{\pi m}}$$

in 2

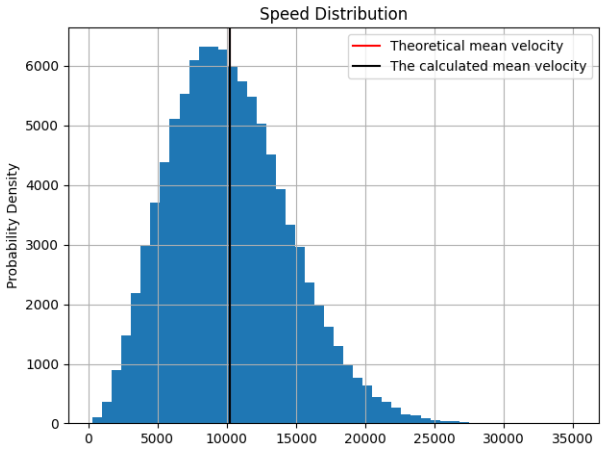


Figure 2: Plot of the actual and theoretical mean velocity with the density distribution

Later, we will compare the theoretical and calculated velocities to check the accuracy of theory against calculations.
The next step is to calculate both theoretical and actual mean kinetic energy of the particles. **Theoretical Kinetic Energy** As deduced in the exercise of the lecture notes "1A", is :

$$\bar{k}_{theoretical} = \frac{3}{2}kT$$

Which is independent of the velocity.

Actual mean kinetic energy was calculated using the classical formula :

$$\bar{k}_{actual} = \frac{1}{2}m\bar{v}^2$$

Here, $\bar{v}$ is the mean velocity of the particles. Both values will be compared. After those calculations , we move to the next step : making an approximated simulation of the gas inside the box.
We start by defining a time interval of 10^{-9}s divided into 1000 steps of iterations.we get :$\Delta t = 10^{-12}$ s. Next, we need to find the particles position at each time step. The relation between the old position and the new position was found by formula :

$$\vec{r}_{t+1} = \vec{r}_t + \vec{v}_t \Delta t$$

We assume that the collision between particles and the walls are elastic. After collision ,particles will gain a new velocity in the x direction with change in velocity : $\Delta v_x = 2vx$ as shown in figure **??**

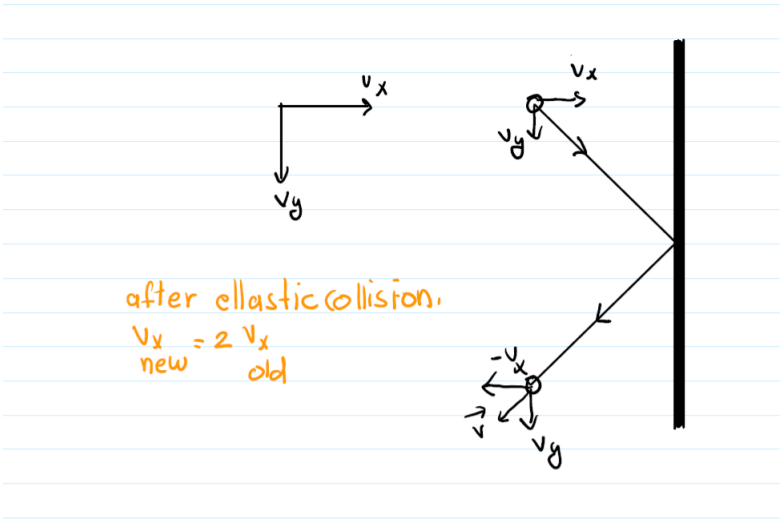


Figure 3: Plot of the elastic collision between the particles and the wall

Finally, the pressure exerted by the $H2$ gas was calculated. For an ideal gas, the pressure is calculated using the ideal gas law:

$$P = nkT$$

where $n = \frac{N}{V}$ is the number density of the gas, k is the Boltzmann constant, and T is the temperature.
Now we need to calculate the actual pressure exerted by the gas, we use the definition of pressure as the force per unit area from the particles in the gas

:

$$P = \frac{F}{A}$$

where$F$ is the total force, A is the Area . As shown in the figure : 4

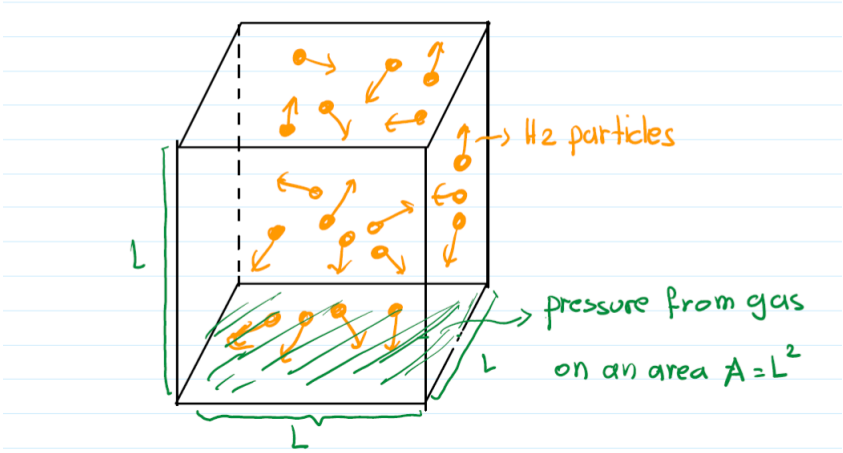


Figure 4: a plot shows the pressure P exerted by a force F on area A

We will assume that the collision with the wall is elastic , that means the total absolute value of the momentum is conserved . After the collision , the particle will have change in momentum : $\Delta P = 2p_x$ A shown in Figure 5

Using Newton second law, the force exerted by the gas particles on the walls of the box is the change in momentum of the particles over time.

$$\vec{F} = \frac{\Delta \vec{p}}{\Delta t} = \frac{\Delta p_x}{\Delta t}$$

After heating up the gas to achieve the optimal energy, next thing we need to do is creating a hole in the box with length $L/2$. The ratio of the area of this hole to the total area of the box is $\frac{1}{24}$. As shown in Figure 6

The escape of particles through this hole would, in a real model, provide the thrust to launch the rocket. According to the conservation of momentum, the escaping particles transfer momentum to the rocket, accelerating it to achieve the escape velocity.

However, for the sake of simplification and avoiding complexity, since we want to compute the fuel needed , we will instead of allowing the particles to escape,will rather choose a model with a" virtual escape" .

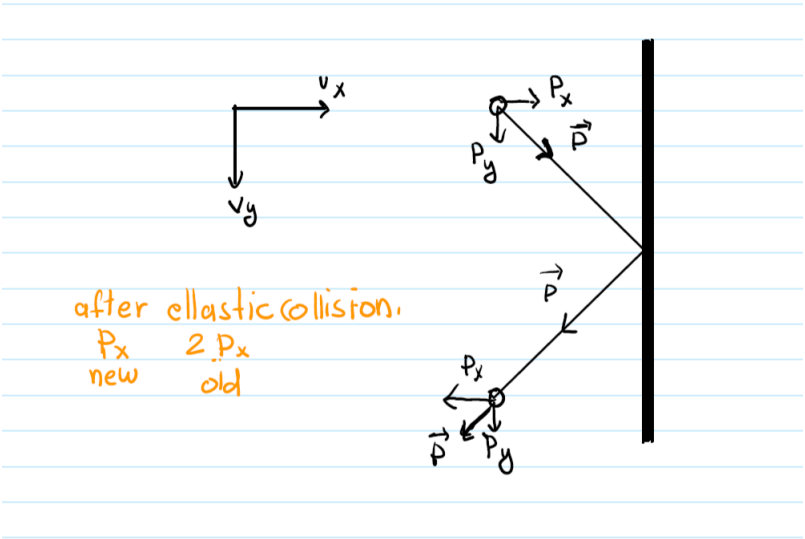


Figure 5: a plot shows the elastic collision with the walls where the absolute value of the momentum is conserved

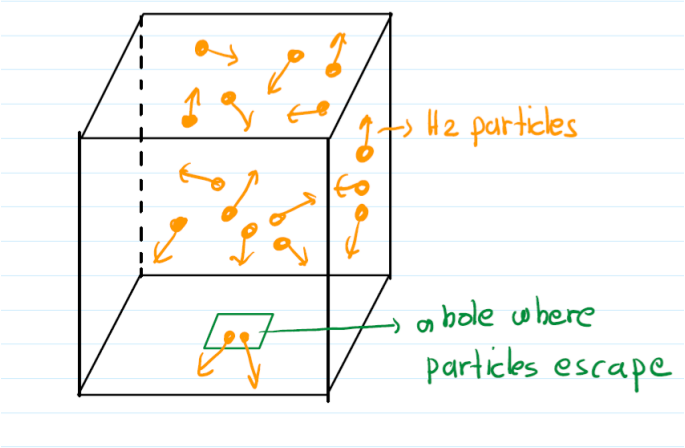


Figure 6: A plot shows a fuel box with a hole through which particles escape

That means we will make the particle after colliding with the walls to bounce back into the box keeping track of the number of particles that would have escaped . We will track the number of particles with position :$|r| >$

$\frac{L}{2}$ which means that the particle is outside the box and then reverse their velocities and sum them up and use that to calculate their momentum like if they virtually escaping by dividing by ratio between the hole area to the box area. $\frac{1}{24}$. The change in momentum in this case will be : $\Delta P = 2mv$. We will later on, find the loss in fuel using the number of virtually escaped particles per hole are:

$$m_{fuel} = \frac{N_{escaped} \times m_{H2}}{A}$$

.

This method allows us to compute the average changing in momentum for all holes and the thrust, without actually losing any particles. In the other approach ,where particles escape and are then replaced by new particles from a fuel tank,would be more complicated. For example, (we need to figure out how to make the particles flow from the chamber to the box in order to keep the density constant).

Using Newton's second law :$a = \frac{F}{m}$, we found $F = \frac{\Delta p}{\Delta t}$ Put them together ,we get:

$$a = \frac{\Delta p}{m_{sat}\Delta t} \tag{1}$$

This is the acceleration per box . where Δp is the change in momentum of the particles, m_{sat} is the mass of the satellite, and Δt is the time interval. To achieve the escape velocity v_{esc} within a time $t = 1200$ seconds (20 minutes), the required acceleration a_{tot} is:

$$v(t) = v_{esc} = a_{tot}t \implies a = \frac{v_{esc}}{t} \tag{2}$$

Here we found the total acceleration and by dividing it by a from (1), we get number of boxes needed to accelerate the satellite .

$$n_{boxes} = \frac{a_{tot}}{a} \tag{3}$$

Lastly,to calculate the mass of fuel needed to accelerate the satellite can be deduced by using the conservation of momentum . The momentum calculated for our " particles virtual escape" is equal to the momentum needed to lift the rocket. we can write

$$P_{particles} = p_{rocket} \implies m_{fuel}\bar{v} = m_{sat}v_{esc} \implies m_{fuel} = \frac{m_{sat}v_{esc}}{\bar{v}} \tag{4}$$

Where $\bar{v} = v_x = \frac{v}{\sqrt{3}}$ is the mean velocity of the particles as we deduced in the lecture notes "1A". This model allows us to count the number of the fuel boxed needed to launch our satellite. The acceleration of the satellite is caused by momentum exchange between the escaped gas particles and the satellite. As particles are escaped at high velocity with a momentum in one direction , the satellite will gain similar momentum in the other direction , causing it to accelerate.

3 Results

- The mass of my planet is $\approx 1.90 \times 10^{25}$Kg or equal to 3.185Earth masses, with radius $R \approx 9386.51$Km.
- The escape velocity was calculated to be : v_{esc} =Km/s.
- The mean value of the position $\approx 2.73 \times 10^{-09}$ m.
- The calculated mean velocity of the particles : $\approx 16.44710229.40$m/s.
- Theoretical mean velocity is: ≈ 10248.37 m/s.
- The calculated mean kinetic energy is $\approx 2.06^{-19}$J.
- The theoretical mean kinetic energy is $\approx 2.07^{-19}$ J.
- the calculated is :$\approx$ 13639.35Pa
- The theoretical pressure is: $\approx$ 13806.49Pa
- the fuel calculated to accelerate the satellite is : 2046.378 Kg
- The theoretical total fuel needed is : 2779.619Kg
- Number of fuel boxes needed $\approx 8.038 \times 10^{12}$ Box.

4 Discussion

from our results:

- The calculated and theoretical mean velocities $\bar{v}$ were in agreement, almost identical.
- The calculated and theoretical mean kinetic energies $\bar{E}_k$ also showed a close match, with a precision of 167/10000.
- The actual and theoretical pressures were consistent, with a precision of 9/1000.
- However, there was a noticeable deviation in the calculated fuel mass compared to the theoretical value. The calculated fuel mass was approximately 37% less than expected.

This deviation can be related to the simplifications we did. We assumed that the pressure and the temperature are both constant. Also we ignored the fuel mass in our simulation. We did not include the effect of gravity and the effect of turbulence of air which has a remarkable impact when moving in a speed of 16.447 Km/s . The simplification we made by assuming that that the particles do not escape,if we made the simulation includes escaping particles, the results would had been more close to reality .

5 Conclusion

We successfully built an acceptable model of a rocket engine. The results were generally convincing, with a deviation observed in the fuel mass calculations. These deviations can be related to the simplifications we made. This is a simple model.It does not represent the reality but it gives an idea about the principle behind launching a rocket. Perhaps , this model would have been more accurate if we have involved the rocket equation in our calculations .

In a real rocket engine, the model would be more complex , considering more variables and factors in the calculations.

References

[1] Frode Kristian Hansen. Lecture Notes 1A *Introduction to Astrophysics,AST2000.*